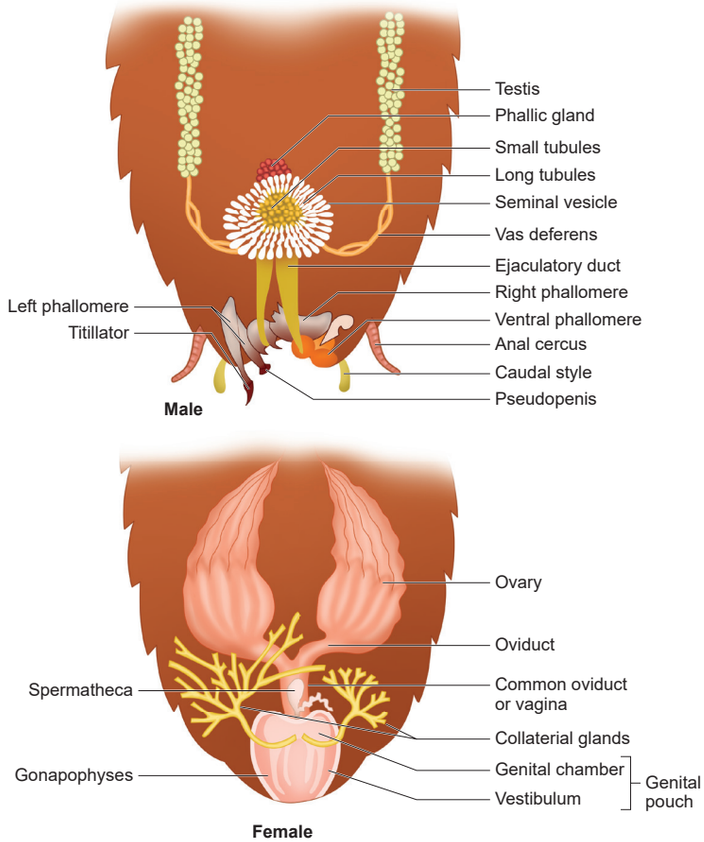


Figure 4.14 *Periplaneta americana reproductive system*

The female reproductive system of cockroach consists of a **pair of ovaries**, **vagina, genital pouch**, **collaterial glands**, **spermathecae** and the external genitalia. A pair of ovaries lies laterally in the 2nd and 6th abdominal segment. Each ovary is formed of a group of eight ovarian tubules or ovarioles, containing a chain of developing ova. The lateral oviducts of each ovary unite into a broad median common oviduct known as vagina, which opens into the genital chamber. The vertical opening of the vagina is the female genital pore. A pair of spermathecae is present in the 6th segment, which opens by a median aperture in the dorsal wall of the genital pouch. During copulation, the ova descend to the genital chamber, where they are fertilized by the sperms. A pair of white and branched **collaterial glands** present behind the ovaries forms a hard egg case called **Ootheca** around the eggs. Genital pouch is formed by the 7th, 8th and 9th abdominal sterna. The genital pouch has two chambers, a genital chamber into which the vagina opens and an oothecal chamber where oothecae are formed. Three pairs of plate like chitinous structures called gonapophyses are present around the female genital aperture. These gonapophyses guide the ova into the ootheca as ovipositors. (Figure 4. 14).

Ootheca is a dark reddish to blackish brown capsule about 12mm long which contains nearly 16 fertilized eggs and dropped or glued to a suitable surface, usually in crack or crevice of high relative humidity near a food source. On an average, each female cockroach produces nearly 15 – 40 oothecae in its life span of about one to two years. The embryonic development occurs in the ootheca, which takes nearly 5 – 13 weeks. The development of cockroach is gradual through **nymphal stages (paurometabolus)**. The nymph resembles the adult and undergoes moulting. The nymph grows by moulting or ecdysis about 13 times to reach the adult form.

Many species of cockroaches are wild. About 30 cockroach species out of 4,600 are associated with human habitats. About four species are well known as pests. They destroy food and contaminate with their offensive odour. The mere presence of cockroaches is a sign of unhygienic condition and they are also known to be carriers of a number of bacterial diseases. The cockroach allergen can cause asthma to sensitive people.

XI Std Zoology Chapter-4 EM.indd 61 30/12/2022 10:36:57